

Class/ Index Number	Centre Number/ 'O' Level Index Number	Name
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新加坡海星中学

MARIS STELLA HIGH SCHOOL

PRELIMINARY EXAMINATION

SECONDARY FOUR

ADDITIONAL MATHEMATICS	4049/2
Paper 2	25 August 2026
Candidates answer on the Question Paper.	2 hours 15 minutes

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.
 Write in dark blue or black pen.
 You may use an HB pencil for any diagrams or graphs.
 Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.
 Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
 The use of an approved scientific calculator is expected, where appropriate.
 You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

For Examiners' Use							
Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
/4	/5	/6	/7	/4	/9	/10	/9
Q9	Q10	Q11	Q12	SUBTOTAL		90	
/8	/9	/9	/10	Statement			
				Presentation			
				Units			
				Rounding Off			

Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1} a^{n-1} b + \binom{n}{2} a^{n-2} b^2 + \dots + \binom{n}{r} a^{n-r} b^r + \dots + b^n,$$

where n is a positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} bc \sin A$$

1 Do not use a calculator in this question.

In triangle ABC , $AB = (6 + \sqrt{5})$ cm and $\angle ABC = 30^\circ$. Given that the triangle has an area of $\frac{1}{4}(32 - 5\sqrt{5})$ cm², find the exact length of BC . Give your answer in the form $(a + b\sqrt{5})$, where a and b are integers. [4]

- 2 In a sports meet, the height, h metres, of a javelin thrown by an athlete is modelled by the formula, $h = -\frac{3}{100}(w^2 - 36w - 76)$, where w is the horizontal distance travelled by the javelin in metres.

- (a) By expressing h in the form $A(w-B)^2 + C$, where A , B and C are constants, find the maximum height reached by the javelin and state the corresponding horizontal distance when this occurs. [3]

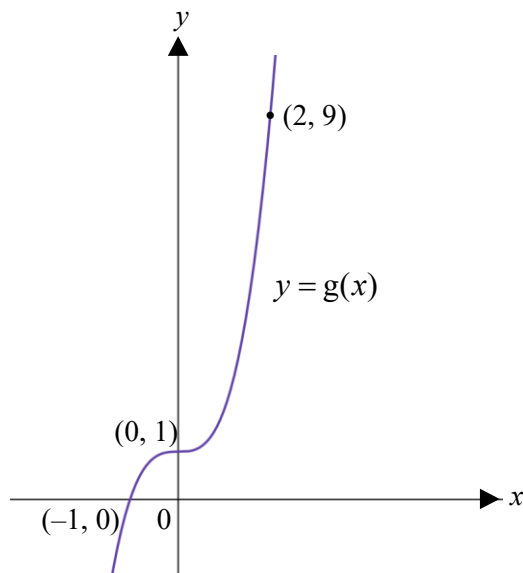
- (b) Find the horizontal distance covered by the javelin before it hits the ground. [2]

- 3 Solve the equation $2 \operatorname{cosec} \theta - 5 \sin \theta = 3 \cot \theta$ for $0^\circ \leq \theta \leq 360^\circ$. [6]

- 4 A curve passes through the point $A(0, -1)$ and the gradient of the curve at this point is parallel to the line $y + 2x = -1$. Given $\frac{d^2y}{dx^2} = 2e^{2x} + 6e^{-2x} + 9\cos 3x$, find y in terms of x . [7]

- 5 (a) Given that $\int_3^{10} f(x) \, dx = 15$ and $f(x)$ lies above the x -axis for $x \geq 3$, find the exact value of $\int_3^7 [5x^2 - 3f(x)] \, dx - \int_7^{10} 3f(x) \, dx$. [3]

- (b) The diagram shows part of the graph of $y = g(x)$.
Given that the curve passes through the points $(0, 1)$, $(-1, 0)$ and $(2, 9)$, find the value of $\int_0^2 y \, dx + \int_1^9 x \, dy$. [1]

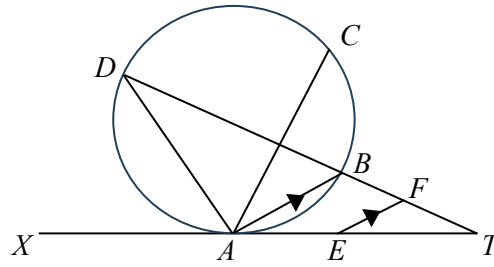


- 6 (a) Find the value of a for which $2x-3$ is a factor of $P(x) = 2x^3 - 5x^2 + ax + 6$, where a is an integer. Hence, determine whether $x+2$ is a factor of $P(x)$. [4]

- (b) Factorise $P(x)$ completely. [2]

- (c) Using your answers in (b), solve the equation $2^{3y+1} - 5(2^{2y}) = 2^y - 6$. [3]

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In the diagram, A , B , C and D lie on a circle. XT is the tangent to the circle at A and the line DB extended meets the tangent at T . Points E and F lie on XAT and DBT respectively. EF is parallel to AB and point E is the mid-point of AT . It is given that AB bisects angle CAT and AD bisects angle CAX .

(a) Prove that $AB = BC$. [2]

(b) Prove that BD is the diameter of the circle. [3]

(c) (i) Prove that triangle TAD is similar to triangle TBA . [3]

(ii) Hence, show that $BT \times DT = 4ET^2$. [2]

8 Given $y = e^{-2x}x^2$,

(a) y is a solution of the equation $e^{2x} \left(4y + 4 \frac{dy}{dx} + \frac{d^2y}{dx^2} \right) = k$, where k is a constant.

Show that $k = 2$.

[6]

- (b) find the range of values of x for which y increases as x increases.

[3]

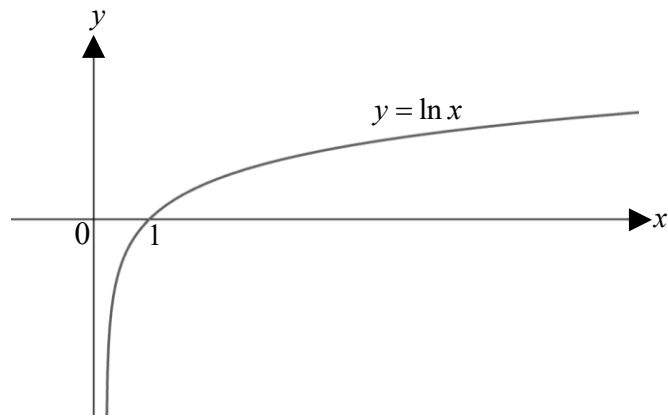
9 It is given that $\lg(2a + b) = \lg a - \lg b$.

(a) (i) Express a in terms of b .

[3]

(ii) State the range of values of a and explain clearly why $0 < b < \frac{1}{2}$.

[2]



- (b) The diagram shows part of the curve $y = \ln x$ for $x > 0$.
Use the graph to explain how you would obtain a graphical solution for the equation $x^2 = e^{4-x}$. [3]

- 10** A particle is travelling in a straight line. Its velocity, v m/s, where t is the time in seconds after passing a fixed point O , is given by $v = 3 - 6\sin 3t$.

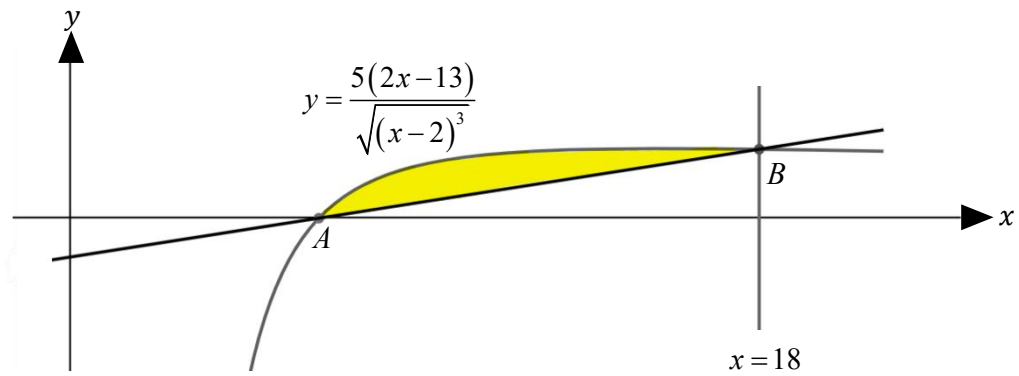
(a) Determine the range of values of v . [1]

(b) Find the exact values of t for the first two times the particle is at instantaneous rest. [3]

- (c) Find the displacement of the particle when $t = \frac{5\pi}{18}$. [3]

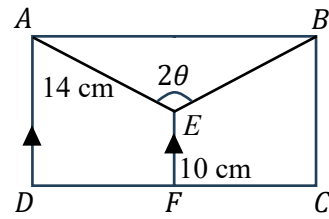
- (d) Find the distance travelled by the particle for the first $\frac{\pi}{2}$ seconds. [2]

11 **(a)** Show that $\frac{d}{dx} \left(\frac{2x+5}{\sqrt{x-2}} \right) = \frac{2x-13}{2\sqrt{(x-2)^3}}$. [3]



- (b) The diagram shows part of the curve $y = \frac{5(2x-13)}{\sqrt{(x-2)^3}}$ where $x > 0$. The curve intersects the x -axis at point A . The line $x = 18$ cuts the curve at point B . Find, using part (a), the area bounded by the curve and line AB . Show all necessary working. Leave your answer correct to 2 decimal places. [6]

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In the diagram, $ABCFD$ shows the design of the front of a handbag. $ABCFD$ is a rectangle, triangle ABE is isosceles with $AE = BE = 14$ cm and $\angle AEB = 2\theta$ radians. EF is parallel to AD and $EF = 10$ cm. The perimeter of the handbag is P cm.

- (a) Show that P can be expressed as $20 + 28\cos\theta + 56\sin\theta$. [2]

- (b) Express P in the form $20 + R\cos(\theta - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$. [3]

- (c) The perimeter of the handbag is lined with a chain of pearls. Given that θ can vary and that the designer paid \$110 for the pearls, find the minimum cost per metre of chained pearls and the corresponding value of θ . [3]

- (d) Explain whether P can be equal to 20 cm. [2]

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MSHS 2026 Prelim AM Paper 2 Answer Key

Qn	Answer	Qn	Answer
1	$(7 - 2\sqrt{5})$ cm	7cii	$\frac{AT}{BT} = \frac{DT}{AT}$ Since $AT=2ET$ (given), $\frac{2ET}{BT} = \frac{DT}{2ET}$ $BT \times DT = 4ET^2 \text{ (shown)}$
2a	$h = -\frac{3}{100}(w-18)^2 + 12$ Maximum height = 12 m, Horizontal distance = 18 m	8a	$k = e^{2x} \left(4y + 4 \frac{dy}{dx} + \frac{d^2y}{dx^2} \right)$ $= e^{2x} \left\{ 4(e^{-2x}x^2) + 4 \left[2e^{-2x}(x - x^2) \right] \right.$ $\left. + 2e^{-2x} - 8e^{-2x}x + 4e^{-2x}x^2 \right\}$ $= 2$
2b	38 m	8b	$0 < x < 1$
3	$\theta = 122.1^\circ, 237.9^\circ$	9ai	$a = \frac{b^2}{1-2b}$
4	$y = \frac{1}{2}e^{2x} + \frac{3}{2}e^{-2x} - \cos 3x - 2$	9b	Draw the line $y = 2 - \frac{1}{2}x$ on the same axes. The x-coordinate of the point of intersection between the two graphs is the solution to the equation.
5a	$481\frac{2}{3}$ or $\frac{1445}{3}$	10a	$-3 \leq v \leq 9$
5b	18	10b	$t = \frac{\pi}{18}$ and $\frac{5\pi}{18}$
6a	Not a factor of $P(x)$	10c	-1.11 m
6b	$P(x) = (2x-3)(x-2)(x+1)$	10d	5.45 m
6c	$y = 0.585$ or 1	11b	7.32 units ²
7a	$\triangle ABC$ is isosceles with base angles $\angle BAC$ and $\angle ACB$, hence $AB = BC$.	12b	$P = 20 + 28\sqrt{5} \cos(\theta - 1.11)$
7b	By property of right angle in a semi-circle, since $\angle DAB = 90^\circ$, BD is a diameter of the circle.	12c	Minimum cost per metre = \$133.16 $\theta = 1.11$
7ci	By AA Similarity Test, $\triangle TAD$ is similar to $\triangle TBA$.	12d	Since $0 < \theta < \frac{\pi}{2}$, hence it is not possible for $P = 20$.